Experiment 20

MySQL String Functions – REPLACE, REPEAT, REVERSE, RIGHT, LEFT, RPAD, LPAD

Aim

To execute MySQL string functions for replacing, repeating, reversing, extracting, and padding strings.

Software Required

MySQL Workbench or MySQL Command Line Client.

SQL Program

Step 1: Create database

```
CREATE DATABASE ShopDB;
USE ShopDB;
```

Step 2: Create table and insert data

```
CREATE TABLE Products (
    id INT PRIMARY KEY,
    product_name VARCHAR(50)
);

INSERT INTO Products VALUES
(1, 'Laptop'),
(2, 'Mobile'),
(3, 'Keyboard');
```

Step 3: Display table

```
SELECT * FROM Products;
```

Queries

REPLACE()

```
SELECT REPLACE('Hello World', 'World', 'MySQL');
```

REPEAT()

```
SELECT REPEAT('Hi ', 3);
```

REVERSE()

```
SELECT REVERSE('MySQL');
```

RIGHT()

```
SELECT RIGHT('Database', 4);
```

LEFT()

```
SELECT LEFT('Database', 4);
```

RPAD()

```
SELECT RPAD('SQL', 6, '*');
```

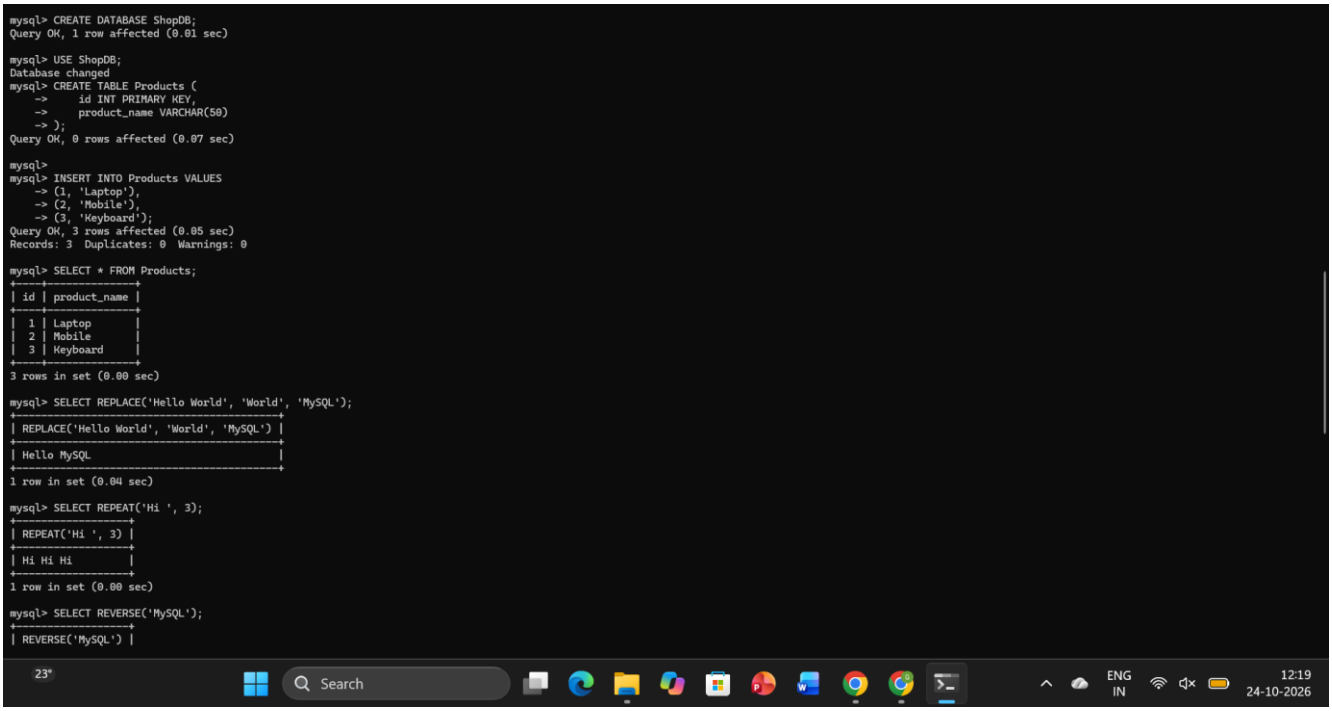
LPAD()

```
SELECT LPAD('SQL', 6, '*');
```

All functions in one query

```
SELECT
  REPLACE('Hello World', 'World', 'MySQL') AS Replace_Result,
  REPEAT('Hi ', 3) AS Repeat_Result,
  REVERSE('MySQL') AS Reverse_Result,
  RIGHT('Database', 4) AS Right_Result,
  LEFT('Database', 4) AS Left_Result,
  RPAD('SQL', 6, '*') AS Rpad_Result,
  LPAD('SQL', 6, '*') AS Lpad_Result;
```

Expected Output:



```
mysql> CREATE DATABASE ShopDB;
Query OK, 1 row affected (0.01 sec)

mysql> USE ShopDB;
Database changed
mysql> CREATE TABLE Products (
    ->     id INT PRIMARY KEY,
    ->     product_name VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.07 sec)

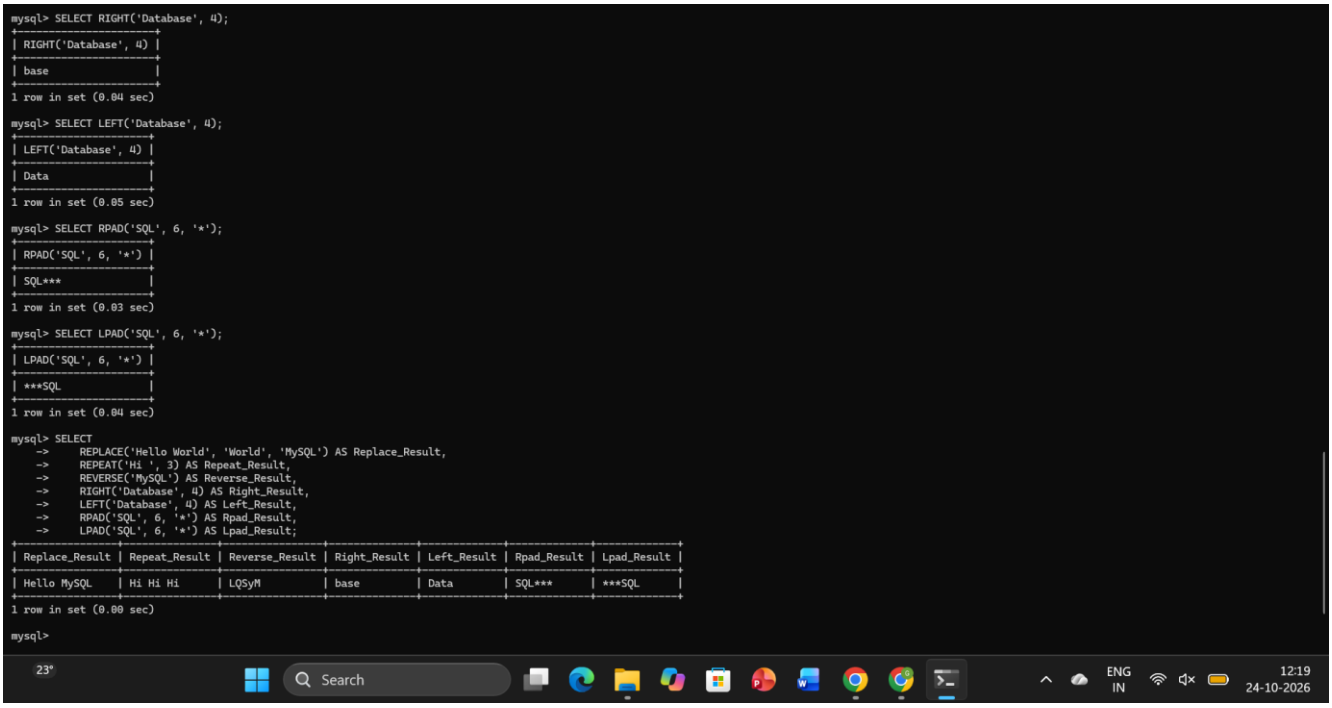
mysql>
mysql> INSERT INTO Products VALUES
    -> (1, 'Laptop'),
    -> (2, 'Mobile'),
    -> (3, 'Keyboard');
Query OK, 3 rows affected (0.05 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Products;
+----+--------------+
| id | product_name |
+----+--------------+
|  1 | Laptop       |
|  2 | Mobile       |
|  3 | Keyboard     |
+----+--------------+
3 rows in set (0.00 sec)

mysql> SELECT REPLACE('Hello World', 'World', 'MySQL');
+-----------------------------------------+
| REPLACE('Hello World', 'World', 'MySQL') |
+-----------------------------------------+
| Hello MySQL                             |
+-----------------------------------------+
1 row in set (0.04 sec)

mysql> SELECT REPEAT('Hi ', 3);
+-------------------+
| REPEAT('Hi ', 3) |
+-------------------+
| Hi Hi Hi         |
+-------------------+
1 row in set (0.00 sec)

mysql> SELECT REVERSE('MySQL');
+------------------+
| REVERSE('MySQL') |
+------------------+
```



```
mysql> SELECT RIGHT('Database', 4);
+----------------------+
| RIGHT('Database', 4) |
+----------------------+
| base                 |
+----------------------+
1 row in set (0.04 sec)

mysql> SELECT LEFT('Database', 4);
+---------------------+
| LEFT('Database', 4) |
+---------------------+
| Data                |
+---------------------+
1 row in set (0.05 sec)

mysql> SELECT RPAD('SQL', 6, '*');
+---------------------+
| RPAD('SQL', 6, '*') |
+---------------------+
| SQL***              |
+---------------------+
1 row in set (0.03 sec)

mysql> SELECT LPAD('SQL', 6, '*');
+---------------------+
| LPAD('SQL', 6, '*') |
+---------------------+
| ***SQL              |
+---------------------+
1 row in set (0.04 sec)

mysql> SELECT
    ->     REPLACE('Hello World', 'World', 'MySQL') AS Replace_Result,
    ->     REPEAT('Hi ', 3) AS Repeat_Result,
    ->     REVERSE('MySQL') AS Reverse_Result,
    ->     RIGHT('Database', 4) AS Right_Result,
    ->     LEFT('Database', 4) AS Left_Result,
    ->     RPAD('SQL', 6, '*') AS Rpad_Result,
    ->     LPAD('SQL', 6, '*') AS Lpad_Result;
+----------------+---------------+----------------+--------------+-------------+-------------+-------------+
| Replace_Result | Repeat_Result | Reverse_Result | Right_Result | Left_Result | Rpad_Result | Lpad_Result |
+----------------+---------------+----------------+--------------+-------------+-------------+-------------+
| Hello MySQL    | Hi Hi Hi      | LQSyM          | base         | Data        | SQL***      | ***SQL      |
+----------------+---------------+----------------+--------------+-------------+-------------+-------------+
1 row in set (0.00 sec)

mysql>
```

Observation

The functions replace, repeat, reverse, extract characters, and pad strings to a specified length.

Result

Thus, REPLACE(), REPEAT(), REVERSE(), RIGHT(), LEFT(), RPAD(), and LPAD() were executed successfully.